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1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s21_3a IS
6      PORT (
7          clk : IN STD_LOGIC;
8          u   : IN STD_LOGIC;
9          y   : OUT STD_LOGIC
10     );
11 END ENTITY;
12
13 ARCHITECTURE behavioral OF s21_3a IS
14     SIGNAL q : STD_LOGIC_VECTOR(4 DOWNTO 0) := "00000";
15     SIGNAL a, b, c, d, e, f, g : STD_LOGIC := '0';
16 BEGIN
17
18     -- combinational circuits
19     a <= u AND g;
20     b <= q(4) AND g;
21     c <= q(3) AND g;
22     d <= q(2) AND g;
23     e <= q(1) AND g;
24     f <= q(4) AND (NOT q(3)) AND q(2) AND (NOT q(1)) AND (NOT q(0));
25     g <= NOT f;
26
27     -- output logic
28     y <= f;
29
30     -- flipflops
31     ff : PROCESS (clk)
32     BEGIN
33         IF rising_edge(clk) THEN
34             q(4) <= a;
35             q(3) <= b;
36             q(2) <= c;
37             q(1) <= d;
38             q(0) <= e;
39         END IF;
40     END PROCESS;
41 END ARCHITECTURE behavioral;

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